

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

26 August 2026

Quick reminder of where yesterday left off: after testing 2028 and 2029 (years the model was never specifically tuned for), a re-tuning test found that using a smaller import amount (20,000 instead of 30,000 megawatts) made the model's hour-to-hour rhythm match Brainpool's real prices much better - so that smaller amount was adopted for both years. Today picked up right there, by asking for a fuller check before settling on that decision.

1. A more complete comparison, including a third option

You asked: "Try 15,000 MW too and compare all three."

Rather than just comparing the newly-adopted 20,000 setting against the original 30,000, an even smaller option (15,000) was added to the comparison too - and this time, the FULL picture was checked, not just the 'ordinary hours only' view that's been used as the main measuring stick throughout this whole project. Conveniently, all three results already existed from an earlier test, so nothing needed to be re-run - just compared properly, side by side.

	15,000 MW	20,000 MW	30,000 MW
Germany2028: hours it ran out of power	49	22	1
Germany2028: average price	69.22	59.30	50.79
Germany2029: hours it ran out of power	116	63	8
Germany2029: average price	94.51	75.51	55.84

Finding: the full picture tells a very different story than the narrow one did.

Looking only at 'ordinary hours' (excluding the rare hours where the model ran completely out of power), the smaller import amounts genuinely did look better - that part of yesterday's finding was real. But once the FULL year is considered, including those 'ran out of power' hours, the picture flips: the smaller the import amount, the WORSE the overall match gets, and the average price ends up badly overshooting Brainpool's real number - for 2029 at the smallest setting, the model's average price came out 53% too high. That's because 'ran out of power' hours aren't just excluded from the count - Brainpool's real forecast barely ever has any, so having 22 to 116 of them a year is itself a sign the overall picture is getting less realistic, not more.

In plain terms: picture judging a bus service only by how punctual it is on the routes it actually completes, while ignoring how often it cancels a route entirely. A service that cancels more routes might look more 'punctual' on paper for the routes it does run - but that's not really a better bus service overall. That's what was happening here: the 'ordinary hours' score looked great, but only because more and more hours were being quietly excluded as 'the model gave up' hours.

2. Deciding what to actually keep

You asked: "Yes, fold it in and keep 30,000 MW as the default."

With the fuller picture in hand, the decision was made to undo yesterday's change and go back to the original 30,000 setting for both 2028 and 2029, rather than keep the smaller one.

Done - reverted cleanly, with everything kept on file.

Both years are back to their original 30,000 setting as the live, default version. Nothing from yesterday's exploration was thrown away - all four tested amounts (15,000, 20,000, 25,000, and 37,650) remain saved separately, so the full evidence trail - including WHY the smaller amount was tried, and WHY it was ultimately not kept - is there for anyone who wants to look at it again later.

This closes the loop honestly: a real lever was found and tested, adopted based on the best information available at the time, and then reconsidered and reversed once a fuller check turned up something the first check had missed - exactly the kind of careful, evidence-first process this whole project has followed throughout.

3. Looking for what else might be fixable

You asked: "What other small adjustments are left to try?"

With the import-amount question settled for now, the natural next move was the same 'where exactly does the mismatch happen' check used many times already this project - but this time run on 2028 and 2029 specifically, since it had never actually been done for those two years before.

A genuine surprise turned up: the weekday/weekend pattern had flipped compared to 2027.

In 2027, weekdays actually matched Brainpool's real prices BETTER than weekends. In both 2028 and 2029, that flips hard - weekdays match much WORSE than weekends, and by a bigger margin than anything seen in 2027. That looked like a real, specific thing worth chasing, rather than random noise.

4. Chasing the weekday puzzle down to a real bug

You asked: "Yes."

Digging into why weekdays specifically were doing worse turned up something concrete: checking the model's own demand numbers by real day of the week showed Fridays coming out unusually LOW and Sundays unusually HIGH - the opposite of what real electricity use actually looks like (real weekday use is highest, real Sunday use is lowest). That's not a modelling choice - that's a sign something is actually broken.

Found it: a genuine calendar bug, quietly present in the project's own 'best' 2027 build since it was built.

The demand shape for 2027 and 2029 is built by borrowing a real year's worth of German electricity usage (2016) and stretching or squeezing it to fit the target year. 2016 has one extra day (a leap year), so that extra day has to be removed to make it fit a normal year. The day that was being removed was February 29th - a day sitting right in the MIDDLE of the year. Removing a day from the middle quietly shifts every single day after it by one day of the week for the rest of the year - so from March onward (10 out of 12 months), every day of the week was mislabelled. This had been sitting undetected in the project's main 2027 build since it was first built.

In plain terms: picture a calendar where someone rips out one page from the middle of the year (say, March 15th) and just glues the rest of the pages back together without renumbering anything. From that point on, every date on the calendar quietly lands on the wrong day of the week for the rest of the year - even though January and February still look completely correct. The mistake is invisible unless you specifically check a date deep into the year, which is exactly what the earlier weekday check happened to do.

5. Fixing it and checking how much it recovers

You asked: "Yes, fix it - rebuild 2027 and 2029, and check the recovery."

The fix: instead of removing the extra day from the MIDDLE of the year (February), it's removed from the very END instead (December 31st) - which doesn't disturb anything before it. This is exactly the trick the 2028 build was already using, for a different reason, and it never had this problem in the first place. Both 2027 and 2029 were rebuilt and re-run with this correction.

	2027: before	2027: after	2029: before
How closely they move together	0.647	0.675	0.349
Average error	28.19	26.91	29.59

A genuinely clean win this time - both numbers improve together, with no catch.

Unlike the import-amount question, this wasn't a trade-off - both years got a real, honest improvement in matching quality AND average error at the same time, with almost no change in how often the model ran out of power. That combination is exactly what you'd expect from fixing an actual mistake, rather than just shuffling numbers around.

You asked: "...and check the ceiling behavior too."

One more thing worth checking: now that the calendar mistake is fixed, does the strange 'smaller import amount works better' pattern from earlier this week still show up, or was that pattern itself somehow caused by the calendar bug?

Checked directly: the two issues are completely separate. Fixing one didn't explain the other.

Re-running the same import-amount comparison on the newly-fixed 2029 model found the exact same pattern as before - smaller import amounts still give noticeably better hour-to-hour matching, at the same cost of far more 'ran out of power' hours. Fixing the calendar bug gave every import-amount setting a small, genuine boost across the board, but it didn't change the underlying shape of that pattern at all. So these are now confirmed to be two real, independent things going on - not one mistake masquerading as two.

Where This Leaves Us

A short, focused day that turned into something bigger - one earlier decision checked more thoroughly and reversed, and then a genuine, previously-undetected bug found and fixed cleanly:

- The 'smaller import amount' idea from yesterday looked good on the narrow 'ordinary hours only' measure, but a fuller check (including the full year and a third option) showed it makes the OVERALL picture worse, not better - more unrealistic 'ran out of power' hours, and a badly overshoot average price.
- The 2028 and 2029 models are back to their original 30,000 setting, matching every other year in this project. The smaller-amount exploration is kept as real, useful, on-file evidence - just not adopted as the answer.
- Checking WHERE 2028 and 2029 specifically disagree with Brainpool (never done before today) turned up

a real, previously undetected calendar bug in how the demand shape is built - present in the project's main 2027 build the whole time, not just the new years. It's now fixed for both 2027 and 2029, with a clean improvement in both matching quality and average error, and no downside.

- Double-checked that this bug wasn't secretly behind the earlier import-amount puzzle - it isn't. Both are real, separate findings, confirmed independently.

The full technical detail behind today's work has also been folded into the main, more detailed progress report (AMIRIS_Germany2027_Progress_Report.pdf) as Phases 27 and 28.